
LinkedIn — <https://www.linkedin.com/in/ritvik-ravi>

Education

Bachelor of Technology — Mechanical Engineering
Indian Institute of Technology Tirupati, India

Expected: May 2027

Relevant Coursework

Mechanical Engineering: Fluid Mechanics, Mechanics of Solids, Thermodynamics, Engineering Dynamics, Material Science & Foundry

Other: Calculus & Differential Equations, Statistics, Mechatronics, Classical Physics, Applied Mechanics Lab, Organizational Behavior

Skills

Programming & Numerical Tools: C, Python, MATLAB

CAD & Simulation: ANSYS Fluent (CFD), ANSYS APDL (Structural FEA), XFOIL, SolidWorks, Fusion 360, AutoCAD

Machine Learning & Data Analysis: Regression and classification models, clustering, feature engineering, model evaluation, applied ML for pattern recognition and decision support

Engineering Focus: Aerodynamics, Structural Analysis, Numerical Modelling, Design Optimization, Systems Thinking

Soft Skills: Technical Communication, Leadership, Cross-functional Collaboration, Project Execution

Projects

LAT Aerospace Wing Optimization — Inter IIT Tech Meet (High Prep) Oct 2025 – Dec 2025
Team Lead

- Led a multidisciplinary team solving a high-prep aerospace problem statement focused on a **propulsion-augmented high-lift wing system**.
- Developed CFD-based aerodynamic models for aero-propulsion integrated wing configurations, including **blown-flap powered lift enhancement**.
- Performed parametric analysis using **ANSYS Fluent, XFOIL, and MATLAB** to study lift, drag, and stall behavior.
- Conducted **structural integrity and safety checks** using FEM (ANSYS APDL) under aerodynamic load cases.
- Extended analysis to include control response under thrust transients, noise considerations, and product-level feasibility.

AVLOK — Fixed-Wing UAV Development

Jan 2025 – Aug 2025

Group Project

- Developing a fixed-wing UAV for long-endurance perimeter surveillance and periodic aerial surveying.
- Led aerodynamic design and sizing, selecting and validating an **Fs63-137(b) airfoil** for endurance-oriented flight.
- Evaluated lift-drag performance, stability characteristics, and mission endurance constraints.

Small-Range Rocket Design and Propulsion Optimization

Nov 2023 – May 2024

Group Project

- Designed and fabricated a solid-propellant model rocket with focus on **nozzle geometry and propellant grain optimization**.
- Studied internal ballistics and thrust performance using analytical calculations and simulation tools.
- Project work was recognized at Tirutsava (annual institute fest) and by the Institute Director.

Star Detection and Constellation Identification App

Jan 2024 – May 2024

Group Project

- Developed an AI/ML-based pipeline to detect stellar centroids from user-captured images.

- Implemented distance-vector matching, nearest-neighbor search, and classification logic for constellation identification.
- Applied machine learning techniques for pattern recognition and noise-robust feature extraction.

Technical Portfolio Website - Engineering & Systems Analysis

Jan 2026 – Present

Independent Project

- Developing a long-term technical portfolio structured as a hybrid of research notes, consulting-style case studies, and engineering essays, with forewords and reviews from leading experts in the field.
- Writing in-depth analyses covering engineering failures, system design trade-offs, aerodynamics, manufacturing considerations, and technology feasibility.
- Emphasis on clarity of reasoning, first-principles analysis, and connecting technical decisions with business and operational implications.

Experience

AlmaNest — CEO & Founder

Aug 2024 – Present

- Founded a student-focused digital platform incubated under IIT Tirupati's incubation cell.
- Led early product design, stakeholder communication, and execution strategy.
- Secured **INR 1.5 lakh** in seed funding; piloted a festival-scale deployment with 600+ users.

Entrepreneurship Cell (E-Cell), IIT Tirupati

Event Coordinator

Sep 2024 – May 2025

Deputy Head

Aug 2025 – Present

- Coordinated logistics and execution for institute-wide hackathons and entrepreneurship events.
- Led cross-functional teams handling sponsorships, operations, and participant experience.
- Promoted to Deputy Head for sustained leadership, reliability, and execution capability.

TEDx IIT Tirupati

Production Lead

Jan 2025 – Feb 2025

Co-Organizer (Lead Organizer)

Dec 2025 – Present

- Led stage design, production logistics, and on-ground execution for TEDx IIT Tirupati 2025.
- Currently responsible for end-to-end planning, speaker curation, team coordination, speaker management, and execution strategy for the upcoming edition.

SPIC MACAY — IIT Tirupati Chapter — Program Coordinator

Jul 2023 – Present

- Managed logistics and programming for classical art performances.
- Served as lead anchor for Kathakali, Odissi, and Veena events.

Statistics Club, IIT Tirupati — Assistant Coordinator

Sep 2023 – Present

- Developed seminar content on probability distributions and statistical inference.
- Conducted a solo session titled “*Lies, Damn Lies, and Statistics*” for 30+ students.

Awards

1. 1st Place — ISRO-SHAR Statewide Space Quiz (World Space Week), 2020
2. 3rd Place — Stratethon Business Case Competition, IIT Dhanbad
3. 2nd Place — Space Quiz, IIT Tirupati & IISER Tirupati (AEthereum Fest)
4. 1st Place — Intra-IIT Cricket Tournament
5. 1st Place (Debate) and 2nd Place (Pop Culture Quiz) — Tirutsava

Participation

- NSSC 2023 — Team Lead (Data Analytics Competition); finalist among **15 national teams**
- Kotler's Maniac, IIT Varanasi — Solo participant; Top **8 of 386**
- Dreams to Dollars, GLIM Chennai — Solo finalist; **3rd place among top 4 teams**